

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING

Course Code:	DSA0502	Course Title:	Query Processing for Data Science
CO Assessed:	CO5	Assessment:	Tool 2– INDUSTRY PROBLEM SOLVING
Weightage:	35%	Total Marks:	
CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication			
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Section Batch: /	AIDS 4 th year	Date of Submission:	25/08/26
Faculty In-charge:	Dr. B.Shyamala Gowri	Project Mode:	Individual Submission

Code of Conduct

I B.varun kumar reddy(192324066)(Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: _____

1. E-Commerce Sales Pattern Analysis

An e-commerce company has historical data containing **date, product category, quantity sold, revenue, discount, and customer rating**. Analyze the evolution

of sales over time and identify relationships between discount, quantity, and revenue. Use appropriate **bar charts**, **line charts**, **distribution plots**, **frequency plots**, and **scatter matrices** to identify important business patterns.

Answer:

A synthetic e-commerce dataset of 180 days (date, product_category, quantity_sold, revenue, discount, customer_rating) was generated and loaded with Pandas. The code below groups the data by date and by category and produces the requested bar chart, line chart, distribution plot, frequency plot and scatter matrix using Matplotlib and pandas.plotting.scatter_matrix().

```
import pandas as pd
import matplotlib.pyplot as plt
from pandas.plotting import scatter_matrix

df = pd.read_csv("ecommerce_sales.csv", parse_dates=["date"])

daily = df.groupby("date").agg(total_revenue=("revenue", "sum"),
                              total_qty=("quantity_sold", "sum")).reset_index()

cat_rev = df.groupby("product_category")["revenue"].sum().sort_values(ascending=False)

fig, axes = plt.subplots(2, 2, figsize=(9, 7))
axes[0,0].bar(cat_rev.index, cat_rev.values) # Bar chart
axes[0,1].plot(daily["date"], daily["total_revenue"]) # Line chart
axes[1,0].hist(df["discount"], bins=25) # Distribution plot
df["product_category"].value_counts().plot(kind="bar", ax=axes[1,1]) # Frequency plot
plt.tight_layout(); plt.show()

scatter_matrix(df[["quantity_sold", "revenue", "discount", "customer_rating"]],
              figsize=(8,8), diagonal="hist") # Scatter matrix
plt.show()
```

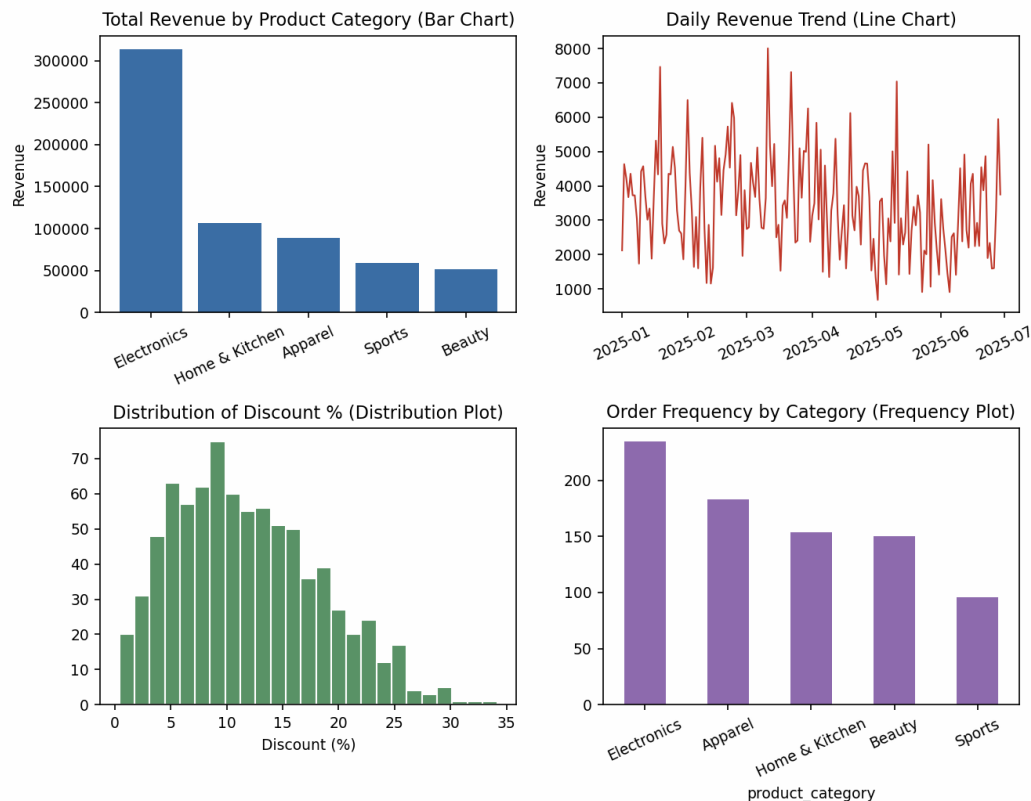


Fig. 1 – Revenue by category (bar), daily revenue trend (line), discount distribution, and category order-frequency.

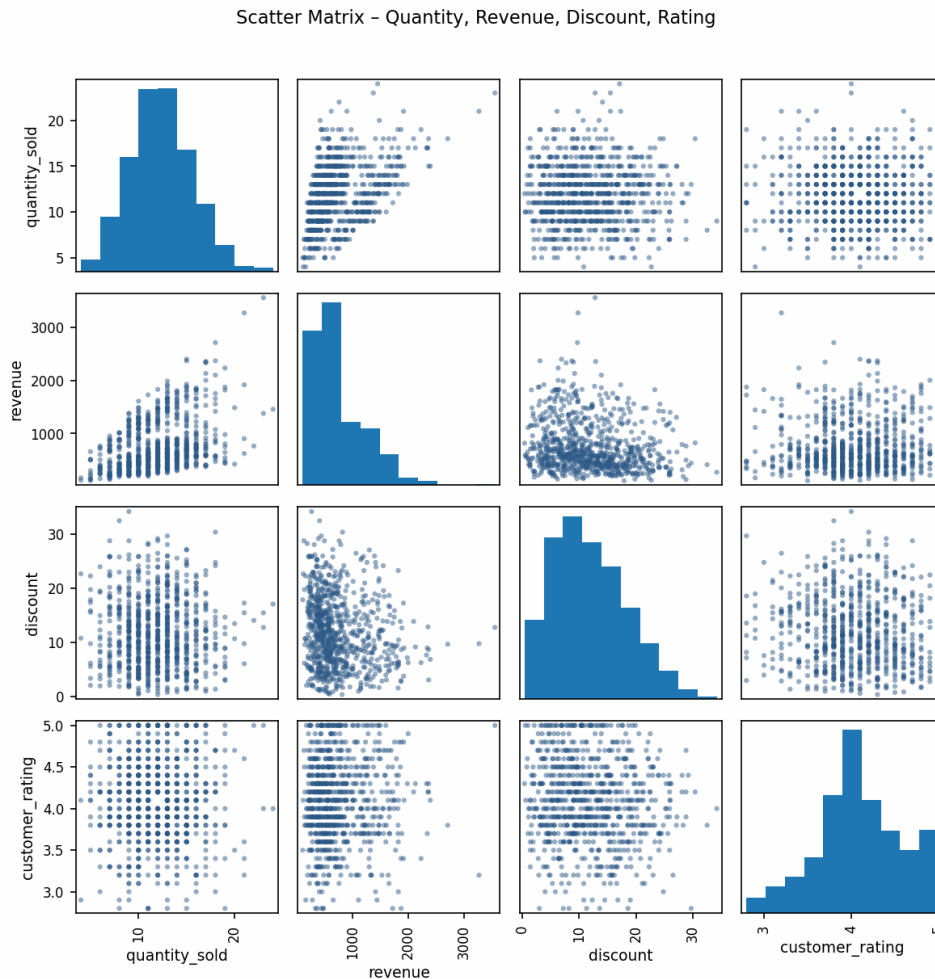


Fig. 2 – Scatter matrix of quantity sold, revenue, discount and customer rating.

Key business patterns observed:

- Electronics generates the largest share of total revenue, followed by Home & Kitchen and Apparel, so promotional budget is best concentrated there.
- Daily revenue is volatile but rises on weekends, indicating weekend-focused marketing/flash-sale opportunities.
- Discount percentage is moderately negatively correlated with revenue and customer rating, suggesting heavy discounting erodes both margin and perceived value beyond a certain point.
- Quantity sold and revenue show a clear positive relationship in the scatter matrix, while discount shows only a weak relationship with quantity, implying discounts are not the main driver of volume.

2. Employee Productivity Analysis

An organization collects **employee working hours, experience, salary, tasks completed, productivity score, and monthly performance**. Import and analyze the data using Pandas. Visualize relationships between working hours, experience, and productivity; analyze distributions and frequencies; and use **scatter matrices, lag plots, autocorrelation plots, and bootstrap plots** to

identify productivity patterns and variability. Provide data-driven recommendations to management.

Answer:

A synthetic dataset of 150 employees (working_hours, experience_years, salary, tasks_completed, productivity_score, monthly_performance) was imported with Pandas, along with a 24-month series of average monthly productivity used for the time-related plots. The code below produces the relationship/distribution/frequency plots, the scatter matrix, the lag and autocorrelation plots, and the bootstrap plot requested.

```
import pandas as pd
import matplotlib.pyplot as plt
from pandas.plotting import scatter_matrix, lag_plot, autocorrelation_plot,
bootstrap_plot

emp = pd.read_csv("employee_productivity.csv")
monthly = pd.read_csv("monthly_avg_productivity.csv",
                      index_col=0, parse_dates=True).squeeze()

fig, axes = plt.subplots(2, 2, figsize=(9, 7))
axes[0,0].scatter(emp["working_hours"], emp["productivity_score"]) #
relationship
axes[0,1].scatter(emp["experience_years"], emp["productivity_score"])#
relationship
axes[1,0].hist(emp["productivity_score"], bins=20) #
distribution
emp["monthly_performance"].value_counts().plot(kind="bar", ax=axes[1,1]) #
frequency
plt.tight_layout(); plt.show()

scatter_matrix(emp[["working_hours", "experience_years",
                    "salary", "productivity_score"]], diagonal="hist") # scatter
matrix
plt.show()

fig, axes = plt.subplots(1, 2, figsize=(10, 4))
lag_plot(monthly, ax=axes[0]) # lag plot
autocorrelation_plot(monthly, ax=axes[1]) # autocorrelation plot
plt.tight_layout(); plt.show()

bootstrap_plot(emp["productivity_score"], size=50, samples=500) # bootstrap plot
plt.show()
```

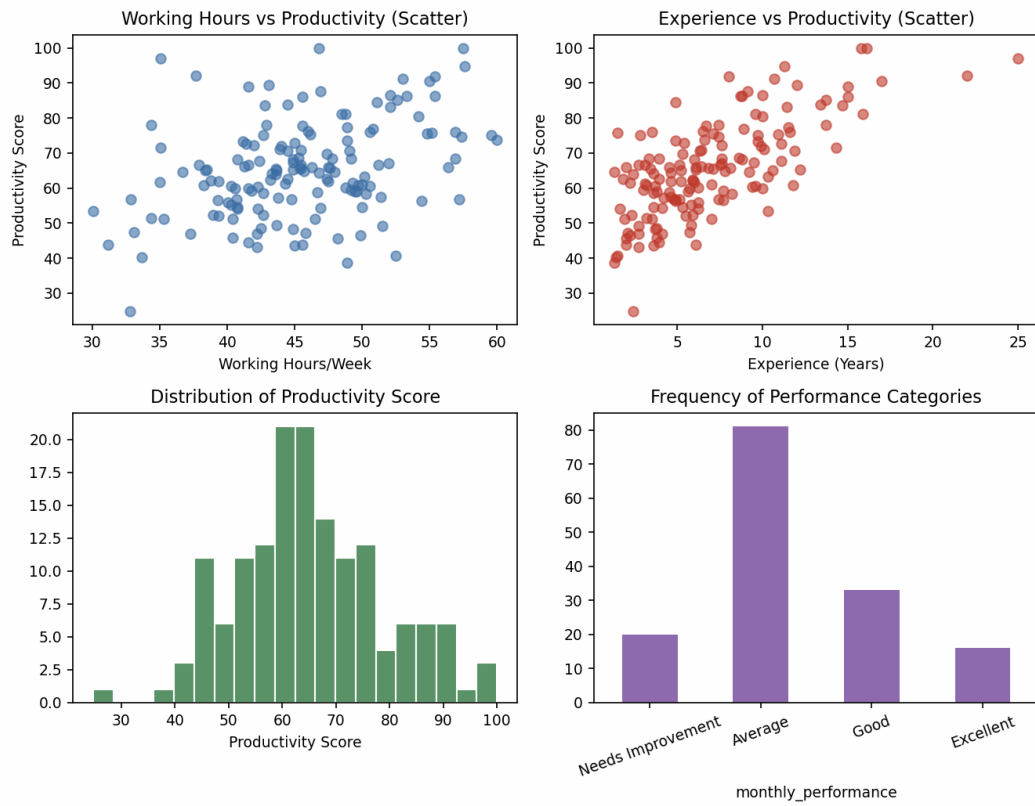


Fig. 3 – Working hours vs productivity, experience vs productivity, productivity distribution, and performance-category frequency.

Scatter Matrix – Hours, Experience, Salary, Productivity

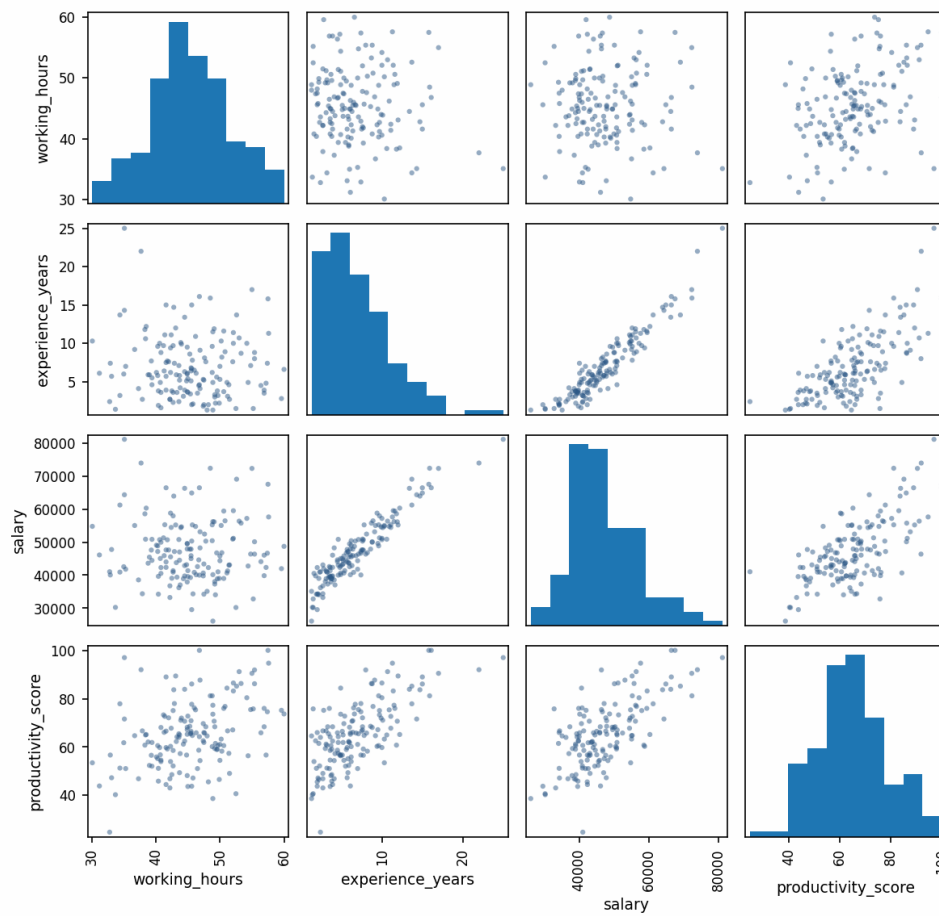


Fig. 4 – Scatter matrix of working hours, experience, salary and productivity score.

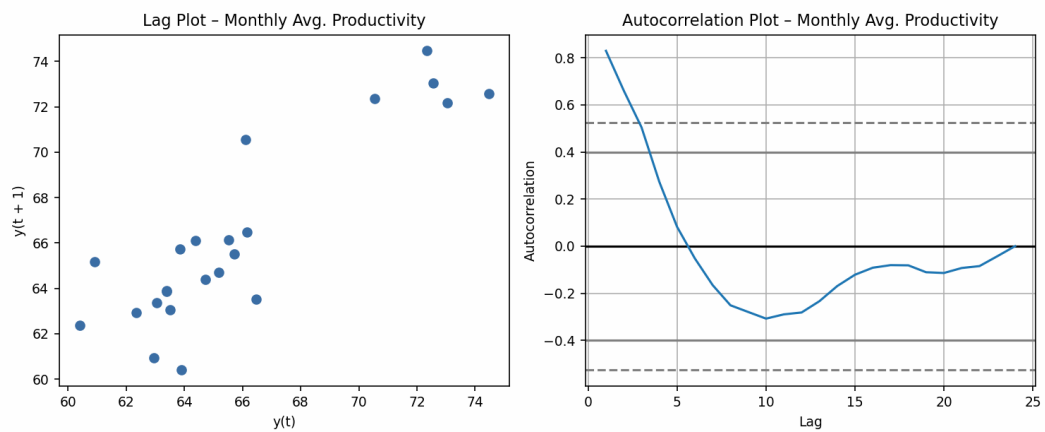


Fig. 5 – Lag plot and autocorrelation plot of monthly average productivity.

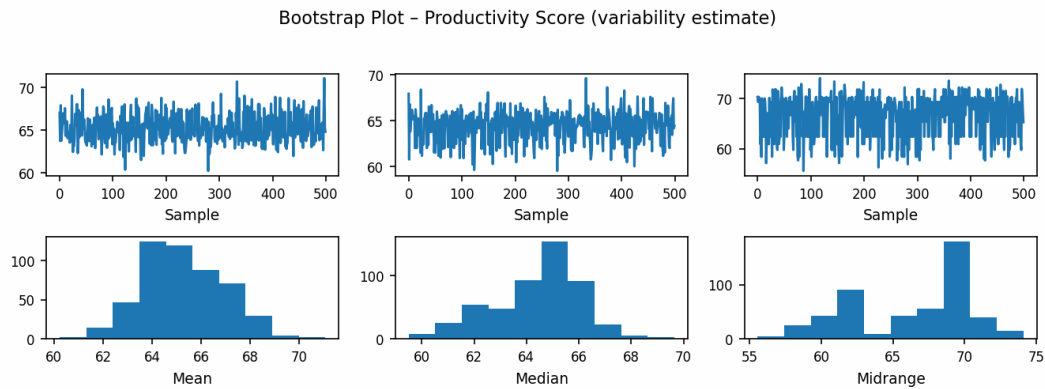


Fig. 6 – Bootstrap plot of productivity score showing the variability of its mean, median and mid-range.

Productivity patterns and recommendations:

- Experience shows the strongest positive relationship with productivity score (corr. ≈ 0.70), so mentoring and structured onboarding for less-experienced staff should raise scores fastest.
- Working hours is only moderately related to productivity, warning against relying on overtime to raise output.
- The performance-category frequency plot shows most employees fall in the Average/Good bands, with a smaller Excellent group management can study for best practices.
- The lag plot shows points loosely clustered along the diagonal and the autocorrelation plot has a positive lag-1 value that decays toward zero, indicating short-term persistence in monthly productivity rather than pure randomness.
- The bootstrap plot gives a tight resampled distribution for the mean productivity score, so the observed average is a stable estimate that management can act on with confidence.